- $E[X] = \sum_x x f(x) = \int_{-\infty}^{\infty} x f(x)\,dx$ · mean exists iff $E[|X|] < \infty$
- $E[r(X)] = \int_{-\infty}^{\infty} r(x) f(x)\,dx = \sum_x r(x) f(x)$
- $Y = r(X_1 \cdots X_n) \Rightarrow E[Y] = \int \cdots \int r(x_1 \cdots x_n) f(x_1 \cdots x_n)\,dx_1 \cdots dx_n$
- $E[aX+b] = aE[X]+b$ · $E[X_1 + \cdots X_n] = E[X_1] + E[X_2] + \cdots + E[X_n]$
- If X_i's are independent $\Rightarrow E[X_1 \cdots X_n] = E[X_1] \cdot E[X_2] \cdots E[X_n]$
- If $P(X \geq a) = 1 \Rightarrow E[X] \geq a$ · If $P(X \leq b) = 1 \Rightarrow E[X] \leq b$
- Jensen's: $g''(x) > 0 \Rightarrow E[g(X)] \geq g(E[X])$
- $\text{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$ · $\sigma = \sqrt{\text{Var}(X)}$
- $\text{Var}(X) = 0$ iff $P(X = c) = 1$ for some constant c
- $\text{Var}(aX+b) = a^2 \text{Var}(X)$ · $\text{Var}(X_1 + \cdots + X_n) = \text{Var}(X_1) + \cdots + \text{Var}(X_n)$
- k^{th} moment: $\mu_k' = E[X^k]$ · k^{th} central moment $\mu_k = E[(X-\mu)^k]$
- Skewness μ_3/σ^3 = measures asymmetry
- MGF: $\psi(t) = E[e^{tx}]$ · $E[X^k] = \psi^{(k)}(0)$ · $Y = aX+b \Rightarrow \psi_Y(t) = e^{bt}\psi_X(at)$
- $Y = X_1 + \cdots + X_n \Rightarrow \psi_Y(t) = \psi_{X_1}(t) \cdots \psi_{X_n}(t)$ · If 2 mgfs are finite and identical in an open interval around $t=0$, their distributions are identical · MSE: $E[(X-d)^2]$ · MAE: $E[|X-d|]$
- M.S.E is minimized when $d = E[X]$, MAE by $d = \text{median}(X)$
- $\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$
- $\text{Cov}(aX+b, cY+d) = ac\,\text{Cov}(X,Y)$ · $\text{Cov}(X+Y, Z) = \text{Cov}(X,Z) + \text{Cov}(Y,Z)$
- $\text{Var}(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2\sum_{i<j} a_i a_j \text{Cov}(X_i, X_j)$
- $\rho(X,Y)$ [correlation] $= \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$ · $\rho(X,Y) \in [-1,1]$
- $[\text{Cov}(X,Y)]^2 \leq \text{Var}(X) \cdot \text{Var}(Y)$ · If $X, Y \to$ independent $\Rightarrow \text{Cov}(X,Y) = 0$
- $|\rho(X,Y)| = 1$ iff $Y = aX+b$ $(a \neq 0)$
- $E[Y|x] \to$ conditional expectation of Y given $X = x$
- $E[Y|x] = \int_{-\infty}^{\infty} y \cdot g_2(y|x)\,dy$ (function of x) · $E[E[Y|X]] = E[Y]$

- $\text{var}(Y|X) = E[(Y - E[Y|X])^2|X]$ • $\text{var}(Y) = E[\text{var}(Y|X)] + \text{var}(E[Y|X])$
- $d(X)$ that minimizes $E[(Y-d(X))^2]$ is $d(X) = E[Y|X]$
- Markov Inequality: $P(X \geq 0) = 1, t > 0 \Rightarrow P(X \geq t) \leq \frac{E[X]}{t}$
- Chebyshev Inequality: $\mu, \sigma^2 \rightarrow$ finite, $t > 0 \Rightarrow P(|X-\mu| \geq t) \leq \sigma^2/t^2$
  $P(|X-\mu| \geq k\sigma) \leq 1/k^2$
- Sample mean $(\bar{X_n})$: $X_1, X_2 .. X_n \rightarrow$ iid; $\mu, \sigma^2 \Rightarrow E[\bar{X_n}] = \mu$
  $\text{var}[\bar{X_n}] = \sigma^2/n$
- Convergence: A sequence of RV's Z_n converge in probability to a constant b if for every $\epsilon > 0$: $\lim_{n\to\infty} P(|Z_n - b| < \epsilon) = 1$
  $(Z_n \xrightarrow{P} b)$
- Law of large no's: $\bar{X_n} \xrightarrow{P} \mu$ (even if variance is ∞)
- If g is continuous at b: $g(Z_n) \xrightarrow{P} g(b)$
- Chernoff bounds: $P(X \geq t) \leq \min_{s>0} e^{-st}\psi(s)$ $\psi(s) = E[e^{sx}]$
- Central limit theorem: distribution of sum/average of large numbers of i.i.d RV's is approximately normal regardless
  Z_n (standardized sample mean) $= \frac{\bar{X_n} - \mu}{\sigma/\sqrt{n}} \Rightarrow Z \sim N(0,1)$
  $\bar{X_n} \sim N(\mu, \sigma^2/n)$; $\Sigma X_i \sim N(n\mu, n\sigma^2)$
- $X \rightarrow$ discrete, $Y \rightarrow$ normal, continuous (approximations)
  - $P(X = k) = P(k - 0.5 < Y < k + 0.5)$
  - $P(a \leq X < b) = P(a - 0.5 < Y < b + 0.5)$
  - $P(X \geq a) = P(Y > a - 0.5)$
  - $P(X \leq b) = P(Y < b + 0.5)$